

JOAN GRAU NOËL

Site Reliability Engineer | Production & Platform Engineering

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PROFILE

Site Reliability Engineer / Software Engineer at Google focused on globally distributed serverless infrastructure, with previous cloud platform engineering roles at Google Cloud and AWS. Focus areas include systems reliability, observability, capacity, efficiency, incident response, and technical leadership. Experience spans C++ debugging, network troubleshooting, and production automation in Go.

EXPERIENCE

Google — Site Reliability Engineer / Software Engineer

April 2022–Present | London, United Kingdom

Work on the Serverless Platform supporting Cloud Run, App Engine, Cloud Functions, and internal Google workloads across many regions. Serve as SRE liaison for the Cloud Run control plane, spanning multiple binaries. The platform spans ingress and egress networking, load balancing, low-latency multi-tenant serving, compute serving and scaling, sandboxed containers, container file serving, control-plane APIs, and databases.

- **Incident leadership:** Led incident response for approximately 20 major customer-facing outages; participated in approximately 100 incidents overall, leading the majority.
- **Distributed tracing:** Led adoption across five software engineering teams and 10+ binaries, connecting request paths across asynchronous Spanner and Pub/Sub boundaries. Reduced p90 time to debug and resolve incidents investigated using tracing by approximately 60%.
- **Go automation:** Built a customer-facing quota broker that uses real-time capacity signals to evaluate resource quota increases. Combined automatic approvals with decision support for manually reviewed requests, reducing associated operational toil for SRE and development teams by approximately 20–30%.
- **C++ reliability safeguards:** Built a presubmit service to detect circular code dependencies and safeguards against draining backends below required capacity, following a major customer-facing outage.
- **Capacity engineering:** Architected and deployed a supply-demand forecasting platform adopted by four partner teams, combining stockout prediction, forecast skew detection, and capacity modelling. Designed a regional scheduling migration to consolidate duplicate capacity buffers.
- **Failover resilience:** Built a continuous indicator of regional N+1 capacity coverage and led remediation of regions below target. Developed a drain simulator to evaluate failover capacity across zone, metro, and disaster-domain failures.
- **Service ownership & leadership:** Own the team's capacity analytics service, setting technical direction with tech leads and collaborating with contributing engineers and multiple partner development teams. Conducted production readiness reviews for autoscaling and deployment features; mentored junior SREs and software engineers.

SYSTEMS ENGINEERING & DEBUGGING

- **C++ diagnostics:** Investigated segmentation faults, memory leaks, race conditions, and performance regressions using heap and core dumps. Added instrumentation to C++ binaries and used distributed tracing to identify cross-service latency regressions.
- **Networking:** Investigated incidents involving eBPF and NAT gateways, using tcpdump packet captures and low-level network debugging. Performed NAT capacity management and planning.
- **Production engineering:** Incident command and operations leadership, postmortems, automated capacity decisions, operational safeguards, production readiness reviews, and mentoring.

EARLIER EXPERIENCE

Google — Technical Solutions Engineer

March 2021–August 2022 | Dublin, Ireland

- Investigated enterprise serverless escalations involving architecture, networking, and performance; collaborated with software and reliability engineers to identify platform issues and recurring failure modes and develop long-term remediation plans.
- Built diagnostic scripts, automation tools, and troubleshooting playbooks to streamline issue reproduction and investigation. Used customer incidents to inform observability improvements and product engineering priorities.

Amazon Web Services — Cloud Support Engineer, DevOps

July 2020–March 2021 | Dublin, Ireland

- Resolved production issues through reproduction, root-cause analysis, and systematic debugging; partnered with service engineering teams on complex customer escalations.
- Advised enterprise customers on scalability, reliability, and operational practices across Elastic Kubernetes Service (EKS), Elastic Container Service (ECS), Lambda, CloudWatch, and AWS CI/CD services.

TECHNICAL SKILLS

Languages: Python, Go, C++ (primary); Java, Kotlin (additional). SQL, Bash/Shell.

Infrastructure: Google Cloud, AWS, Kubernetes, Docker, distributed systems, Protocol Buffers.

Reliability: Distributed tracing, SLIs/SLOs, monitoring and alerting, capacity forecasting, N+k redundancy, regional scheduling, and failover modelling.

Data analytics: SQL, data pipelines, forecasting, regression analysis.

EDUCATION & PROJECT

Universitat Pompeu Fabra — Audiovisual Systems Engineering

2014–2018 | Barcelona, Spain

Thesis — Audio source separation: Adapted a CNN-based PyTorch architecture for monaural audio source separation and applied data augmentation to improve model output. [Source code and thesis.](#)